Here the ratios between successive terms are equal. Therefore the sequence $\frac{1}{2}$, 1, 2, 4, ... is a Geometric Progression with common ratio $r = 2$.

(iii) 5, 25, 50, 75,...

$\frac{t_2}{t_1} = \frac{25}{5} = 5$; $\frac{t_3}{t_2} = \frac{50}{25} = 2$; $\frac{t_4}{t_3} = \frac{75}{50} = \frac{3}{2}$

Since the ratios between successive terms are not equal, the sequence 5, 25, 50, 75,... is not a Geometric Progression.

Thinking Corner

Is the sequence $2, 2^2, 2^{2^2}, 2^{2^{2^2}}, \ldots$ is a G.P. ?

Example 2.41 Find the geometric progression whose first term and common ratios are given by (i) $a = -7$, $r = 6$ (ii) $a = 256$, $r = 0.5$

Solution (i) The general form of Geometric progression is a, ar, ar^2,...

$a = -7$, $ar = -7 \times 6 = -42$, $ar^2 = -7 \times 6^2 = -252$

Therefore the required Geometric Progression is $-7, -42, -252,...$

(ii) The general form of Geometric progression is a, ar, ar^2,...

$a = 256$, $ar = 256 \times 0.5 = 128$, $ar^2 = 256 \times (0.5)^2 = 64$

Therefore the required Geometric progression is 256, 128, 64,....

Progress Check

1. If first term $= a$, common ratio $= r$, then find the value of t_9 and t_{27}.
2. In a G.P. if $t_1 = \frac{1}{5}$ and $t_2 = \frac{1}{25}$ then the common ratio is _____.

Example 2.42 Find the 8th term of the G.P. 9, 3, 1,...

Solution To find the 8th term we have to use the nth term formula $t_n = ar^{n-1}$

First term $a = 9$, Common ratio $r = \frac{t_2}{t_1} = \frac{3}{9} = \frac{1}{3}$

$$t_8 = 9 \times \left(\frac{1}{3}\right)^{8-1} = 9 \times \left(\frac{1}{3}\right)^7 = \frac{1}{243}$$

Therefore the 8th term of the G.P. is $\frac{1}{243}$.

Example 2.43 In a Geometric progression, the 4th term is $\frac{8}{9}$ and the 7th term is $\frac{64}{243}$. Find the Geometric Progression.

Solution 4th term, $t_4 = \frac{8}{9} \Rightarrow ar^3 = \frac{8}{9}$ ...(1)

7th term, $t_7 = \frac{64}{243} \Rightarrow ar^6 = \frac{64}{243}$ ...(2)

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